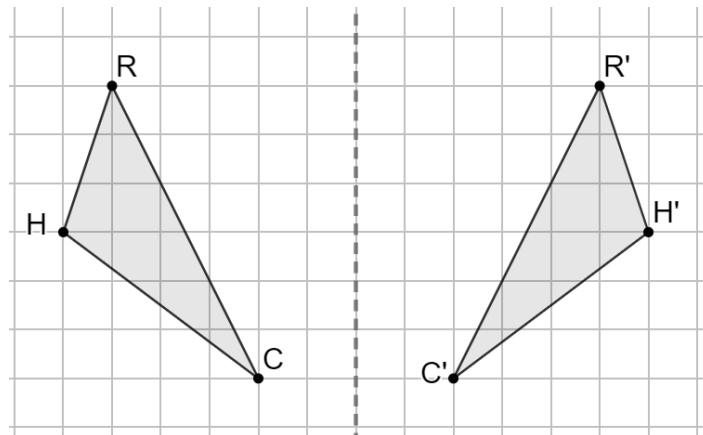


Geometry
Unit 4
Lesson 6 Practice

Name _____
Date _____
Period _____

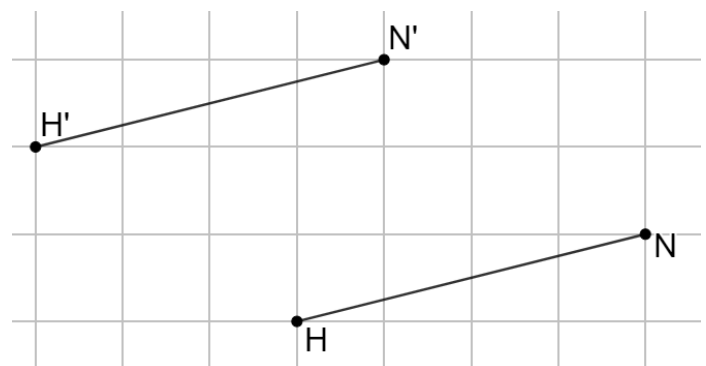
Triangles RHC and $R'H'C'$ are shown on grid paper.



1. Explain what method(s) can be used to determine if each angle of $\triangle RHC$ has the same measure as $\triangle R'H'C'$.
2. Explain what method(s) can be used to determine if each side of $\triangle RHC$ has the same length as $\triangle R'H'C'$.
3. Explain why the transformation of $\triangle RHC$ shown on the grid can be classified as isometry.

Line segments HN and $H'N'$ are shown on the grid.

4. Show the mapping of point H to H' .
5. Describe the transformation applied to $\overline{HN}$ that resulted in $\overline{H'N'}$.
6. Complete the statement.



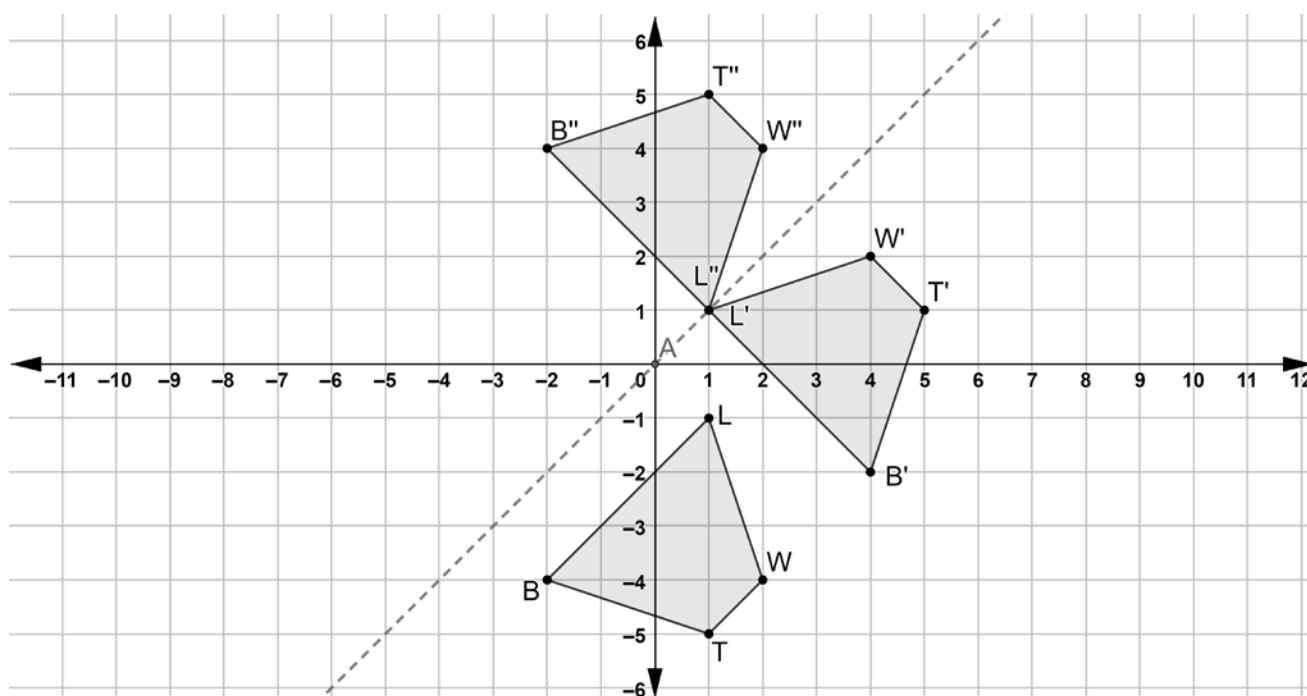
- ☐ Each point on
- ☐ Only the endpoints of

$\overline{HN}$ can be mapped to

- ☐ a corresponding point on
- ☐ the endpoints of

$\overline{H'N'}$.

Quadrilaterals $LWTB$, $L'W'T'B'$, and $L''W''T''B''$ are shown on the coordinate grid.



7. Describe what movements should be done with a patty paper copy of quadrilateral $LWTB$ so that the copy of $LWTB$ lays on top of $L'W'T'B'$.
8. Describe what movements should be done with a patty paper copy of quadrilateral $L'W'T'B'$ so that the copy of $L'W'T'B'$ lays on top of $L''W''T''B''$.
9. Complete the statements.

The transformations used in both sequences

☐ are
☐ are not

examples of isometry. All of the points of

☐ $L'W'T'B'$
☐ $L''W''T''B''$
☐ $L'W'T'B'$ and $L''W''T''B''$

can be mapped to all of the points of $LWTB$.
10. Use the definition of congruence in terms of rigid motions to explain how both sequences of transformations for quadrilateral $LWTB$ resulted in $LWTB \cong L''W''T''B''$.